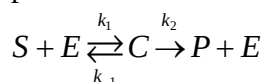


PC5215, Numerical Recipes with Applications, Lab 5

(due Friday 12 Nov 2010)

Enzymes are proteins that act as catalysts for reactions. This means that they only act in an intermediate way, and the final product finds the enzyme molecules unchanged (and looking for some more action). Enzymes are important for most biochemical reactions and are also of importance to industry, especially in food processing. A simple scheme for an enzymatic reaction is that the enzyme reacts with a species called a substrate to form a complex; a further reaction results in a desired product and free enzyme. Symbolically, this is



Note that C can decay in two ways, the new product $P+E$, or the starting chemicals $S + E$. k_1 , k_{-1} and k_2 are the reaction rates.

- (a) Write down a system of (first order) differential equations for the kinetics of the various concentrations (using the law of mass action, check up a physical chemistry book, if you don't know the "law").
- (b) At $t = 0$, suppose that $C = P = 0$ (We'll use the name of the chemical species for its concentration as well). Show that

$$E(t) + C(t) = E(0)$$

Eliminate E , leaving a coupled pair of equations for C and S .

- (c) Because of the result in (b), the equations are essentially reduced to two degrees of freedom. Solve the equations numerically, plot $S(t)$, $C(t)$, $E(t)$ and $P(t)$ as a function of t for some typical initial conditions and reaction rate constants k_i . It's up to you to set these parameters and look for interesting features of the solutions. Discuss the general features you have found. Specify the parameters and initial conditions.
- (d) Draw the phase space diagram in the (S,C) space (parametric curves $[S(t), C(t)]$) and discuss its general features.

You may use NR's `odeint()` or other routines (such as Matlab or Mathematica) to solve the equations. You do need a way of plotting your results. Explore your possibilities. Please indicate your parameters and initial conditions clearly, so that I can check your answer.